set-windows-and-door: ran 3 times; moved openings_set 3, pieces_not_priced 3
frame-loft: ran 7 times; moved joists_set 7, pieces_not_priced 7
lay-subfloor: ran 1 times; moved deck_laid 1, pieces_not_priced 1
install-boiler: ran 1 times; moved boiler_installed 1

ASSUMPTIONS, EVERY SETTING WHOSE REASON SAYS IT IS ASSUMED

floor.thickness_mm = 114 mm; reason: assumed: a 4-1/2 in (114 mm) SIP floor panel, the wall standard's thickness, research row sip.prs610_panel_thickness; Jim's floor build-up gives it
weather.u_value = 2.8 w_per_m2_k; reason: assumed: typical double glazing, not chosen by Jim
weather.solar_heat_gain_coefficient = 0.6 ratio; reason: assumed: typical double glazing, not chosen by Jim
weather.inside_celsius = 20 celsius; reason: assumed design day
weather.outside_celsius = 0 celsius; reason: assumed design day
weather.sun_watt_per_square_metre = 500 w_per_m2; reason: assumed clear-sky peak; the solver applies it to every side
finish.mean_reflectance = 0.5 ratio; reason: assumed mid-grey interior; Jim's wall finish gives it
finish.sky_angle_degrees = 65 degree; reason: assumed open sky; the site gives it
envelope.wall_r = 20 r_value; reason: assumed R-20 for a SIP wall (ft2 F h per BTU); the SIP thickness gives it
envelope.roof_r = 30 r_value; reason: assumed R-30; the roof build-up gives it
envelope.floor_r = 20 r_value; reason: assumed R-20; the floor build-up gives it
envelope.door_r = 5 r_value; reason: assumed R-5 for an insulated door, not checked against a source
heating.boiler_efficiency = 0.7 ratio; reason: assumed 70 percent; sources run 55 to 80 percent
loft_floor.thickness_mm = 18 mm; reason: assumed: 23/32 in (18 mm) combination subfloor, the IRC subfloor table's panel for a 24 in span, research row bom.subfloor_irc; at 16 in joist spacing a thinner panel may do
joists.member_depth_mm = 140 mm; reason: assumed 2x6; the beam solver says whether it is enough
joists.spacing_mm = 406 mm; reason: 16 in on centre, assumed
header.plies = 2 count; reason: assumed: two plies side by side, 3-1/2 in of LVL or 3 in of 2x lumber, fit inside the SIP's 3-5/8 in core (sip.prs610_core_thickness)
roof_loads.roof_dead_load_psf = 15 psf; reason: assumed: 15 psf, the roof dead load Boise's roof-only header look-up assumes, research row gl.hdr.vl_lookup_roof_only; the roof build-up gives it
roof_loads.deflection_limit_ratio = 240 ratio; reason: assumed: span over 240, the live-load deflection limit of Boise's header tables, research row gl.hdr.vl_limits, applied to the whole load
roof_loads.roof_bears_on = south, north compass_sides; reason: assumed: the roof's 8 ft south hangover can only be carried by members running north-south across the south wall, so the roof bears on the south and north walls

UNMEASURED PIECES

none

PROJECT	SHEET	SHEET NUMBER	DATE
Small house on posts, from Jim's pencil sketch	**Design summary**	S-1 (page 3 of 3)	2026-10-20
CONFIG COMMIT	ENGINE PIN	SCALE	SOURCE
eddda50	we a29552e; rendered with we 976c052	not to scale	we geometry